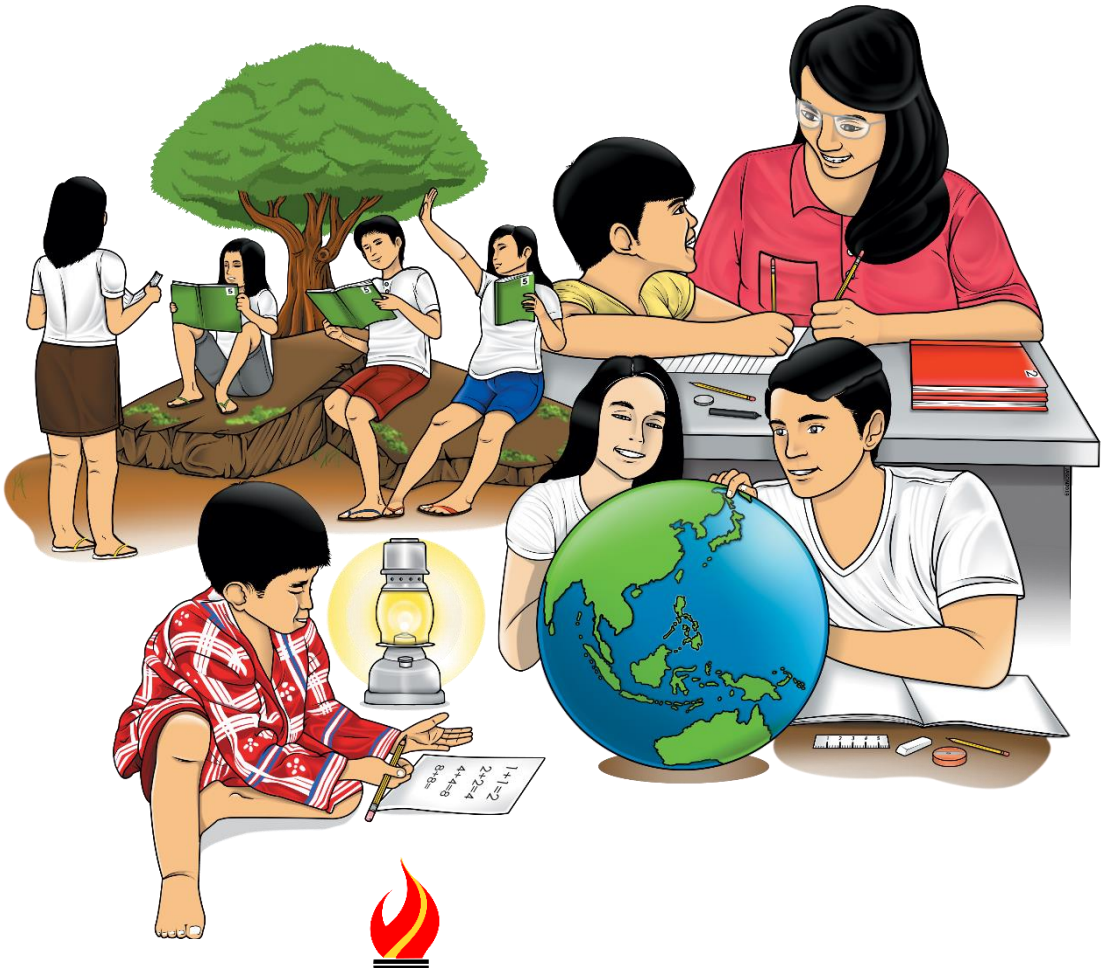
Senior High School

DepED
DEPARTMENT OF EDUCATION

General Biology 1

Quarter 1



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What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of General Biology 1. The scope of this module allows for it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module focuses on the lesson:

- Lesson 1 – Cell Theory
- Lesson 2 – Cell Structures and Functions

After going through this module, you are expected to:

1. Explain the postulates of the cell theory
2. Identify the different cell organelles
3. Describe the functions of the cell organelles.

What I Know

Let me know how far your knowledge on this topic is. Begin your journey by answering the Matching Type test below.

Directions: Match the cell structures and its functions. Write the letter of the correct answer.

A	B
___1. Cytoplasm	a. it is for collection, packaging, and distribution
___2. Mitochondria	b. site of protein synthesis
___3. Ribosomes	c. holds the organelles in place
___4. Lysosomes	d. also known as digestive sacs
___5. Golgi apparatus	e. gives energy to the cell
___6. Organelles	f. brain of the cell
___7. Cell membrane	g. separates cells from other cells
___8. Nucleus	h. basic unit of life
___9. Microscope	i. tiny organs that perform important life processes in the cell
___10. Cell	j. an instrument that makes smaller objects to larger

Lesson 1 — Cell Theory

All living things are either characterized as being composed of cells or as cells themselves-membranous compartments filled with organic molecules in such a way that they can be autonomous self-replicating units of life.

At the end of this module, you are expected to understand the factors that make planet earth habitable.

Leaning Competencies:

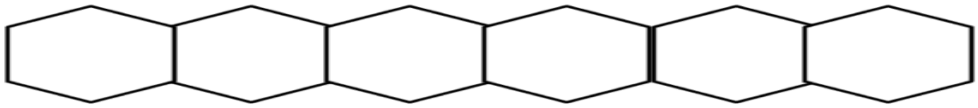
- Explain the postulates of cell theory **(STEM_BIO11/12-Ia-c-1)**
- Describe the structure and function of major and subcellular organelles **(STEM_BIO11/12-Ia-c-2)**

What's In

Did you see how microscopically the cork cell looks like? They were also the cells that Robert Hooke found after cutting out a slice of bark from an old oak tree in 1665. Robert Hooke's history of discovering the presence of cells led other cytologists to unlock the secret of the cell world.

What's New

Directions: Using the interconnected hexagon below, create a puzzle timeline of the history of the cell theory.



What is It

Cell theory is one of biology's founding principles. This theory suggests that cells establish every living thing as its basic unit of structure.

The cell was first described by Robert Hooke, an English experimental scientist. He used this simple microscope to investigate very thin cork slices, a material made from dead plant tissue. He observed that the cork has a honeycomb of tiny compartments; the compartments are empty because the cork cells were dead. In the cork *cellulae*, he named the compartments, and coined the word cell. Nevertheless, Hooke did not know the real structure and function of cells, and his findings did not give any hint of the nucleus and other organelles that were present in the most living cells.

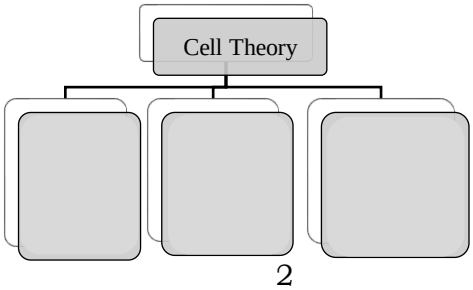
In 1670, a Dutch scientist Anton van Leeuwenhoek used a microscope with 300 times magnification and discovered bacteria and protozoa that he collectively called animalcules.

Biologist refused to recognize the importance of cells for nearly two centuries. Then, in 1839, plant and animal tissues were examined by the German biologists Matthias Schleiden (a botanist) and Theodor Schwann (a zoologist) and confirmed that all living things consist of cells. Rudolf Virchow concluded in 1858 that all cells come from pre-existing cells, thereby completing the classical theory of cells. The belief that cells make up all living things is called the Cell Theory. The cell theory embodies three principles in its modern form.

1. The cell is the fundamental unit of structure and functions in an organism.
2. All living creatures consist of one or more cells.
3. Cells arise from previously existing cells through cell division.

What's More

Directions: Supply the missing principles in the concept map below to complete the cell theory. Write your answer on the space provided).



What I Have Learned

Directions: Explain each principle of cell theory? Write your answer on the space provided.

_____.

What I Can Do

Directions: Create a comic strip which could be accurate, funny, simple and explanatory about cell theory.

Lesson 2 — Cell Structures and Functions

All living organisms are composed of cells. A cell is a small structure enclosed in the membrane, filled with an aqueous solution where organelles and other subcellular structures are located. Cells are formed a shaped differently.

What's In

Directions: Arrange and rewrite the following sentences below in proper order to complete the basic components of cell theory.

1.Living are composed of more things one or all cells.

_____

2.Unit cell is the life of basic.

_____

3.Pre-existing cells arise from all cells.

_____

4. All living things is the belief that make up the cells called cell theory.

_____

5.Cells that live is made up of everything.

_____

What's New

Directions: Follow the instructions below.

1. Click this link https://www.slideshare.net/KhanSaif2/cell-structure-and-function-110716980
2. Using the information from the presentation, complete the table below.
3. Illustrate and describe at least five (5) organelles.

Name of Organelles	Illustration	Description

What is It

All cells have small organs which are called organelles. Organelles are specialized cell parts which have a specific function to do. The first part is the cell membrane that separates the inside parts of the cell from its surroundings. A jelly-like fluid, cytoplasm in which the organelles are suspended. The nucleus serves as the control center and stored the genetic information of the cell. Another structure is ribosomes which synthesize proteins. These free-floating ribosomes can be in the cytoplasm or attached to the endoplasmic reticulum known as the rough endoplasmic reticulum (RER) while smooth endoplasmic reticulum (SER) has no ribosomes on it. This organelle served as the passageway for protein synthesis. As proteins move, into the packaging center of the cell, Golgi apparatus, some of the materials will be used by the cell while others stored in a sac-like structure called vacuoles. However, another material that is not needed by the cells will be stored in the lysosomes which serve as the garbage collector that take in or worn out the cell parts. Once the material circulates in the cell organelles including the mitochondria it will now convert food into energy.

What's More

Activity: 3-D Plant and Animal Cell Models

Procedures:

1. Choose between plant and animal cell.
2. Gather recyclable materials which can be seen in your house to be used in your 3-D model.
3. Construct a 3-D cell model from the available materials and properly label it.
4. Take note that cell model must be 3-D, it cannot be flat like a poster.

Guided Question:

1. How would you describe your 3-D model?

What I Have Learned

Directions: Answer the following questions and write it on the space provided.

1. What kind of organisms are composed of cells? _____
2. Is the statement, vacuoles lie inside the cell, true or false? _____
3. What is the main function of cytoskeleton? _____
4. What is the main function of chlorophyll? _____
5. Is the sentence true or false? Some cell structures are specific to either plant or animal cells? _____
6. What is the function of cell wall? _____
7. What important molecules do the ribosomes produce? _____

8-9. What is the difference between rough endoplasmic reticulum and smooth endoplasmic reticulum?

_____

10. Is the sentence true or false? Ribosomes contain protein _____

What I Can Do

What would happen if one of the cell organelles stopped working properly?

_____

_____.

Assessment

Directions: Choose the letter of the BEST answer. Write your answer on the space provided before the number.

_____1. Where is the location and function of nucleus?
A. It is close to mitochondria and the control center of the cell
B. It is close to mitochondria and makes proteins.
C. It is in the middle of the cell and store fats.
D. It is in the middle of the cell and breaks food.

_____2. What organelle is associated with energy conversions?
A. cytoplasm B. mitochondria C. nucleus D. ribosomes

_____3. What is the function and location of lysosome?
A. It break down foods and located in the cytoplasm.
B. It makes proteins and located in the middle of the cell.
C. It transport materials inside the cell and located in the cytoplasm.
D. It stores genetic information and located in the middle of the cell.

_____4. What is the function and location of chloroplast?
A. It converts light energy into useful energy in the body and close to rough endoplasmic reticulum.
B. It contains the green pigment of plant and in the cytoplasm.
C. It allows the cell division and close to nucleus.
D. It support the cell and located outside the cytoplasm.

_____5. What is the function and location of the Golgi apparatus?
A. It is the packaging center of the cell and located in the cytoplasm.
B. It stores the genetic information of the cell and in the middle of the cell.
C. It synthesizes proteins and located through-out the cell.
D. It makes proteins and through-out the cell.

Additional Activities

Directions: Create your own cell analogy (a comparison between two things, typically for the purpose of explanation or clarification.) using the city 's various department**.** Match the part of city with at least five (5) parts of the cell according to its functions. No. 1 is already provided as an example.

Cell City Analogy

Parts of the cell	**Functions**	**Part of city (City's various department)**
Ex. 1. Nucleus	Control center	**Mayor's Office**

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the nature of General Biology 1. Since you are now familiar with cell organelles and their functions, you will now learn how to distinguish the differences between the two main cell types, you read them can be changed to correspond with the textbook you are now using.

The module focuses on the lesson:

- Lesson 1 – Prokaryotic vs. Eukaryotic Cells
- Lesson 2 – Types of Cells

After going through this module, you are expected to:

1. Distinguish prokaryotic and eukaryotic cells according to their distinguishing features.
2. Classify different cell types (of plant/animal tissues) and specify the function(s) of each.

What I Know

Directions: Fill in the KWHL Chart below to assess your prior knowledge and understanding of the topic, Prokaryotes vs. Eukaryotes and Cell Types.

K	**W**	**H**	**L**
What do I Know?	**What do I want to find out?**	**How can I found what I learn?**	**What did I learn?**

Lesson 3 — Prokaryote vs Eukaryote

Do all cells have the same structure? Let us have an analogy; an efficiency apartment is a one-room apartment. This one room is where you sleep, eat, shower, and entertain your guests. It all happens in one room. It is a simple way of living in a small space. On the other hand, a mansion is a large, complex living space with many separate rooms. There are rooms for cooking, eating, sleeping, bathing, reading, watching TV, entertaining guests, exercising, and storage. The rooms in a mansion are constructed for the specific things you would like to be able to do. You can live in simple efficiency or complexity. Just like the cells, those are as simple as a one-room efficiency apartment or as complex as a mansion.

After going through this module, you are expected to:

1) Distinguish prokaryotic and eukaryotic cells according to their distinguishing features. **(STEM_BIO11/12-Ia-c-3)**
2) Classify different cell types (of plant/animal tissues) and specify the functions of each. **(STEM_BIO11/12-Ia-c-4)**

What I Need to Know

Organelles are the biggest difference between bacteria and cells that make up the human body. The number one biggest difference between the bacteria in your body and the cells making up your body are these tiny cellular components called organelles. You have learned a lot about organelles in other lessons without knowing it. Organelles are simply membrane-bound compartments within a cell, such as the nucleus, mitochondria, chloroplasts, Golgi, and endoplasmic reticulum.

What's In

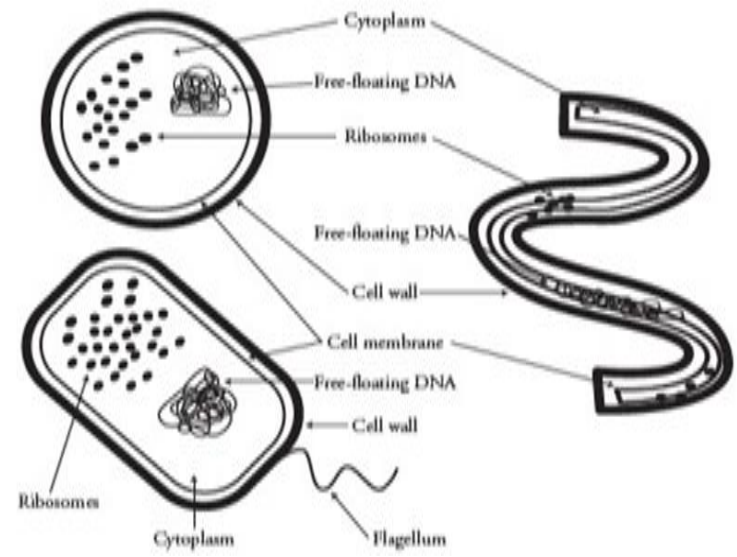
Your body is composed of trillions of cells that make up different organ and other parts. Your body is also a 'home sweet home' for several bacteria. But do not be afraid – not all bacteria harm us. You and the bacteria have a few things in common. They both contain DNA as a heritable genetic material, and they can reproduce.

What's New

Activity- Pro Or Eu

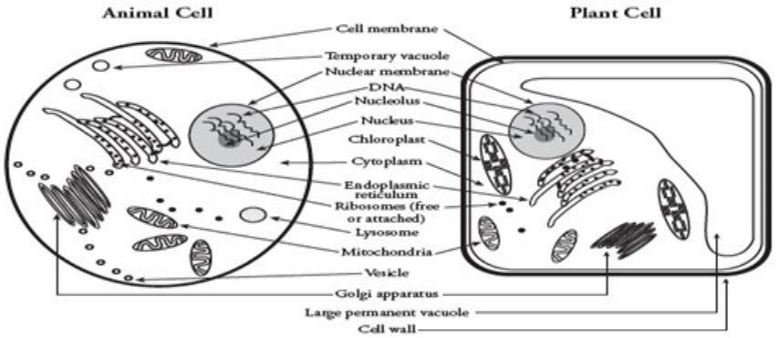
Directions: Distinguish prokaryotic and eukaryotic cells according to their distinguishing features, then answer the given questions.

Model 1: Three Types of Bacterial Cells



1. What is represented by the small dots found in each of the bacteria cells? _____
2. What is the name of the outermost layer that forms a boundary around the outside of each nucleus? _____

Model 2: – Animal and Plant Cells

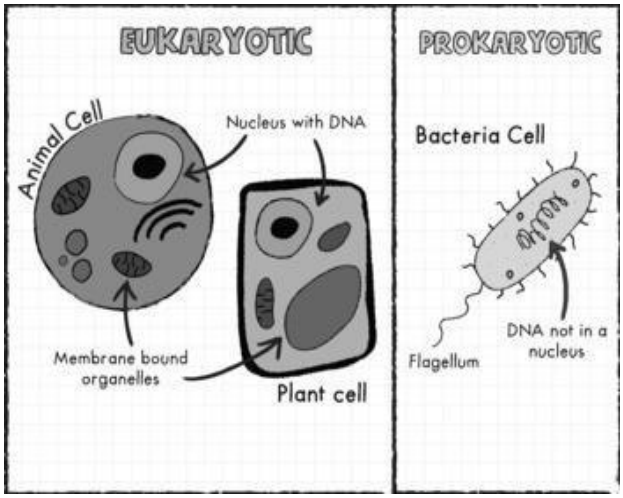


3. Do both cells in Model 1 and 2 have nucleus? _____
4. What is difference between the nuclear membrane in Models 1 and 2? _____
5. Which of the cells in Models 1 and 2 are more complex? List at least three supporting reasons for your choice. _____

What is It

You are a eukaryote. **Eukaryotic cells** contain membrane-bound organelles, including a nucleus. Eukaryotes can be single-celled or multi-celled, such as you, plants, fungi, insects, and me.

Bacteria is an example of prokaryote. **Prokaryotic cells** do not contain a nucleus or any other membrane-bound organelle. Prokaryotes include two groups: bacteria and another group called archaea.



https://www.teacherspayteachers.com/Product/Prokaryotic-Vs-

What's More

Directions: Write YES or NO in the box for each cell.

	Bacterial Cell	**Animal Cell**	**Plant Cell**
Cell Membrane			
Ribosome			
Cytoplasm			
Mitochondria			
Nucleolus			
Nucleus			
DNA			
Cell Wall			
Prokaryotic			
Eukaryotic			

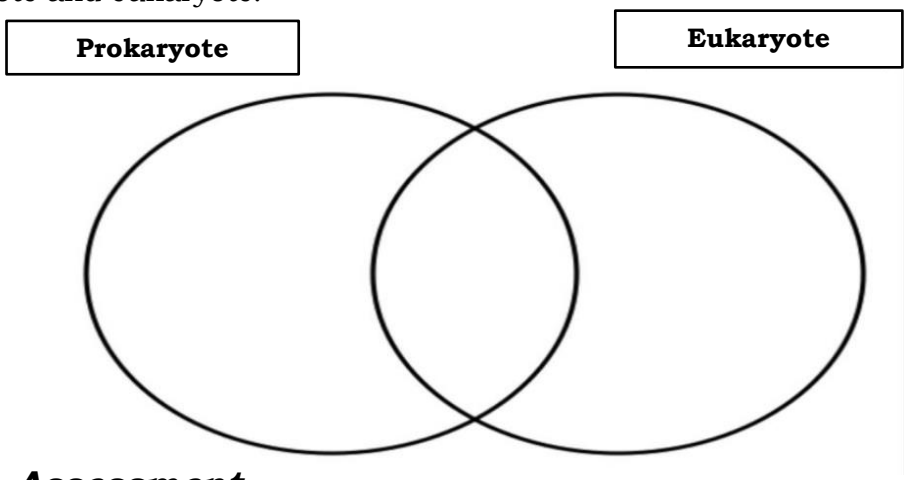
What I Have Learned

Directions: Explain your answer.
Discuss the relationship between a mansion and a cell.

_____
_____.

What I Can Do

Complete the Venn diagram by supplying the similarities and differences between prokaryote and eukaryote.



Assessment

POST-TEST

Directions: Encircle the letter of the best answer.

1. Which of the following statements is True?
   A. Eukaryotic cells do not have a nucleus, Prokaryotic cells do.
   B. Eukaryotic cells are simple, Prokaryotic cells are complex.
   C. Prokaryotic cells are bigger than Eukaryotic cells.
   D. Prokaryotic cells came before Eukaryotic cells.
2. Which of the following is **NOT** a difference between Prokaryotic and Eukaryotic Cells?
   A. The complexity of the cells C. The age of the cells
   B. The size of the cells D. Having a nucleus
3. A cell has a nucleus, is complex and large, and has membrane bound organelles. What type of cell is this?
   A. Bacteria C. Prokaryotic
   B. Eukaryotic D. There is not enough information.
4. A cell has circular DNA, is small and simple, and no nucleus or membrane bound organelles. What type of cell is this?
   A. Animal B. Plant C. Eukaryotic D. Prokaryotic
5. A scientist finds a cell that is very old and very simple. This cell is probably a ___
   A. eukaryotic cell C. plant cell
   B. new type of cell D. Prokaryotic cell

Additional Activities

Make a research about common bacteria that causes skin itchiness brought by flood.

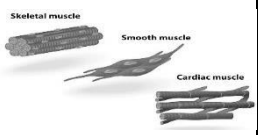
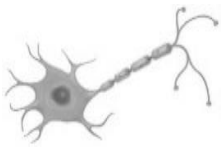
Lesson 4	Specialized Cells

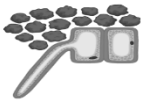
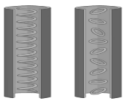
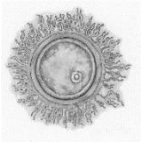
Unicellular – single-celled – organisms such as bacteria need to perform all the functions necessary for life on their own. In contrast, multicellular organisms can divide up these functions among different types of cells. This is known as cell specialization.

Your body contains over 200 different types of specialized cells. Each type is adapted to do a particular job well and has developed special features to do it. Just as you cannot use a pair of scissors as a hammer, you cannot replace a bone cell with a muscle cell and expect to get the same result.

What I Need to Know

Organelles are the biggest difference between bacteria and cells that make up the human body. The number one biggest difference between the bacteria in your body and the cells making up your body are these tiny cellular components called organelles. You have learned a lot about organelles in other lessons without knowing it. Organelles are simply membrane-bound compartments within a cell, such as the nucleus, mitochondria, chloroplasts, Golgi, and endoplasmic reticulum.

Cell Type	What do I look like?	What makes me special?
1. Muscle Cell 	✓These cells are long and can contract quickly. ✓They contain lots of mitochondria to produce energy required for movement.	✓ Have bands of protein and are arranged in fibers to help them run along the length of the muscles.
2. Nerve Cell 	✓Very thin and thread-like with lots of nerve endings (dendrites) at either end.	✓ Transmit electrical impulses around the body, and very connected to other nerve cells sending messages to the brain, so the body responds in a particular way.
3. Palisade Cell	✓Located near the top of any leaf that is easily recognizable because it is long, thin and packed full of chloroplast.	✓ Its chloroplasts, which are full of a pigment called chlorophyll is responsible for photosynthesis.
4. Phloem	✓Found inside plants which is very long and join end to end and have only very few subcellular structures.	✓ Transport food (mainly dissolved sugars) around the plant and therefore needs to make hollow tubes.

5. Root Hair Cells 	✓Attached to plant roots which is long and thin with a large surface area.	✓ Long and thin and have a large surface area to help with the absorption of water and minerals.
6. Sperm Cell	✓Made in the testes and can be found in semen that is long and thin and have tail.	✓ Can swim with the help of the tail that helps it to find egg cells to fertilize it.
7. Xylem Cells 	✓Found inside plants that is a dead cell joins together end to end with other dead cells.	✓ Carry water and mineral ions from the roots to the leaves and stem.
8.Ciliated Epithelial Cell	✓Found lining certain tubes in the body, including the trachea and bronchi, which lead to the lungs that have cilia(tiny hair-like structures) on the surface to sweep particles away.	✓ Protect the lungs from microorganisms and dust by moving them up the airway.
9. Egg Cell 	✓Made in the ovaries of a female that is the largest, roundest human cell, surrounded by a protective layer of cells that provides nutrients.	✓ Contain half of the genetic information (DNA) needed to make a baby.
10.Red Blood Cells	✓ Made in the bone marrow and can be found in the blood vessels that are a biconcave disc shape, which gives a large surface area and allows it to travel easily through blood vessels.	✓ Travel inside the blood vessels carrying oxygen to all cells so it can be used for respiration.
11.White Blood Cells	✓ Usually made in the bone marrow and can be found in the blood vessels which are typical animal cell with a lobed nucleus.	✓ Help fight pathogens which cause disease

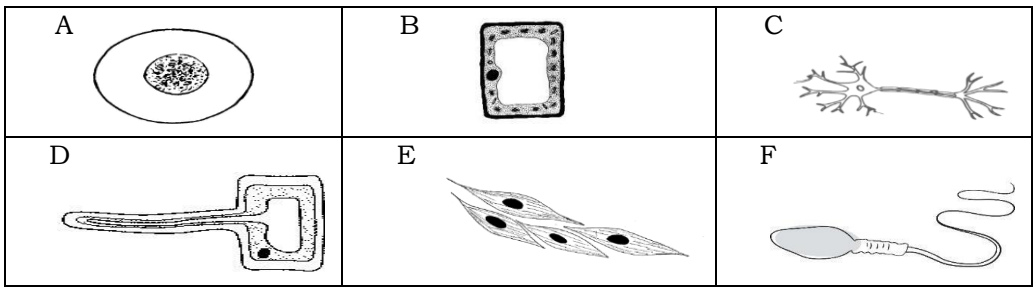
What's In

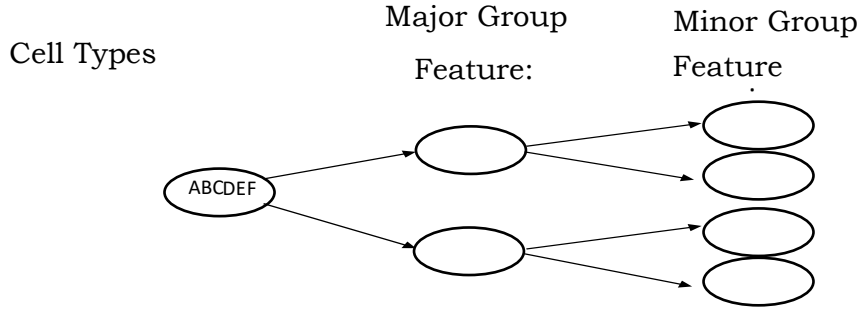
Specialized cells perform specialized functions in multicellular organisms. Groups of specialized cells cooperate to form a tissue, such as a muscle. Different tissues are in turn grouped together to form larger functional units, called organs. Each type of cell, tissue, and organ has a distinct structure and set of functions that serve the organism.

What's New

Directions: Classify the cells into groups based on their important features.

- The diagrams show six different types of cells.
- Classify the six cell types into two major groups by writing their letter in the major group circles, then into minor groups.





What is It

The human body is made up of microscopic cells. These building blocks of life combine and work in harmony. While many cells make up simple body parts, such as tissues, some have complex and specialized tasks. These **specialized cells** are specially designed to perform the functions for which they are intended. Each of these cell types are formed and operate differently, ensuring that the cell can carry out the necessary body function. Specialized cells vary widely with respect to shape, size, number of organelles and types of organelles.

What's More

Activity-Who Am I?

Directions: Identify what type of cell is being described.

1. I have a special rod-shaped extension that is sensitive to light. Who am I?
2. I connect with other cells of my kind to form message-carrying networks. Who am I?
3. I can fight infections by changing shape to swallow and eat bacteria. Who am I?

What I Have Learned

Directions: Explain your answer.

What is the importance of specialized cells in our body?

_____

_____

What I Can Do

Directions: Complete the table below.

Type of Cell	Illustration	Function

Assessment

POST-TEST

Directions: Read each question carefully and encircle the letter of the best answer.

1. Which of the following types of cells sends messages throughout the body in the form of electrical impulse?
   A. muscle cell B. nerve cell C. skin cell D. sperm cell
2. Which of the following types of cells carries oxygen throughout the body?
   A. gamete B. platelet C. red blood cell D. white blood cell
3. Which of the following types of cells fights infection?
   A. plasma B. platelet C. red blood cell D. white blood cell
4. Which of the following types of cells is found in females and is needed for reproduction?
   A. egg cell B. muscle cell C. platelet D. sperm cell
5. How are nerve cells specialized?
   A. Instead of a nucleus, they have a lot of hemoglobin to help them carry oxygen.
   B. They have long connections that transmit messages quickly.
   C. They have a large amount of cytoplasm.
   D. None of these.

Additional Activities

Directions: Imagine you are a type of cell in the human body. Your task is to research your cell type and then prepare a resumé so that you can apply for a job that suits your skills. Organize your resumé as follows: Use a separate paper, it can be computerized or handwritten.

- ✓ Name: state which type of specialized cell you are.
- ✓ Address: identify where in the body you can be found.
- ✓ Place of birth: identify where in the body you were formed.
- ✓ Appearance: provide two images – a drawing and a microscope image – and label your main features, such as your shape, organelles, and any special extensions.
- ✓ Skills: list your main roles in the body and explain how your various features allow you to perform them.
- ✓ Clubs and organizations: describe your role in forming tissues and organs and

your associations with any other specialized cell types.

- ✓ Closing statement: write a paragraph to describe the type of job you are looking for and highlight how your skills would suit this job.

  Some examples of cells you might choose to be nerve cell (neuron), red blood cell (erythrocyte), sperm cell, egg cell, photoreceptor cell, taste bud cell, heart muscle cell, fat cell (adipocyte), macrophage.

Lesson 5 — Cell Modifications

Cells undergo modification or differentiation for them to perform their function effectively and efficiently. Microvilli, cilia, stereocilia, and flagella are examples of those cell modifications.

Learning Competencies

- Describe some cell modifications that lead to adaptation to carry out specialized functions (e.g., microvilli, root hair) **(STEM_BIO11/12-Ia-c-5)**
- Characterize the phases of the cell cycle and their control points **(STEM_BIO11/12-Id-f-6)**

What I Know

ENTRY SLIP: Fill up this entry slip before you proceed with the activity proper.

3 Things that you want to know about the lesson:
2 Things that you are expecting about the lesson:
1 Thing that you are quite uncertain about the lesson:

What's In

Arrange the following jumbled letters to reveal the hidden word.

_____1. (ILLIV) Finger-like projections that arise from the epithelial layer in some organs that help increase surface area allowing for faster and more efficient absorption.

_____2. (GELLAFLA) Microscopic hair-like organelle used by microorganisms for movement.

_____3. (TIHGT IONJUNCT) Act as barriers that regulate the movement of water and solutes between epithelial layers and prevent leakage of extracellular fluid.

_____4. (EDSOMSOEM) Found beneath the zonula adherens and throughout the cell, which are made of protein filaments with filaments of the

adjacent cell that creates dense intermediate line between the cells and act like spot welds between adjacent epithelial cells.

_____5. (ILICA) Motile, involves in movement and appear as short hair-like structures formed from microtubules.

What's New

Cell modifications are specialization or modifications that resulted from reacquisition after cell division which allow efficient function that promotes survival of the organism. Examples of plant and animal cell modifications are microvilli, cilia, stereocilia, and flagella.

What is It

Apical modifications found on the apical surface of the cell. Villi are finger-like projections that arise from the epithelial layer in some organs. They help to increase surface area allowing for faster and more efficient absorption. Microvilli are also called brush/striated border, are finger-like cytoplasmic extensions of the apical surface which increase surface area for absorption just as found in small intestines.

Stereocilia are long, non-motile microvilli that function in increasing absorption and are found in sensory cells in ear and male reproductive tract. Cilia are motile, involve in movement and appear as short hair-like structures formed from microtubules meanwhile, flagella are microscopic hair-like organelle used by cells and microorganisms for movement. Aside from usage for locomotion, flagella used to increase reproduction rates, sense changes in the environment (temperature or pH disturbances) and may even be used as a secretory organelle.

Basal modification, a cell modification found on the basal surface of the cell. Basal surface contacts the basal lamina since it contains receptors for blood borne factors such as hormones. Desmosomes (macula adherens) are found beneath the zonula adherens and throughout the cell, which are made of protein filaments with filaments of the adjacent cell that creates dense intermediate line between the cells and act like spot welds between adjacent epithelial cells. Additional functions are pinning adjacent cells together, ensuring that cells in organs and tissues that stretch, such as skin and cardiac muscle remain connected in an unbroken sheet. Hemidesmosomes are like desmosomes in terms of function, however, they attach the epithelial cell to the basement membrane rather than to an adjacent cell. Basal infoldings, often found in epithelium that transport fluid such as kidney.

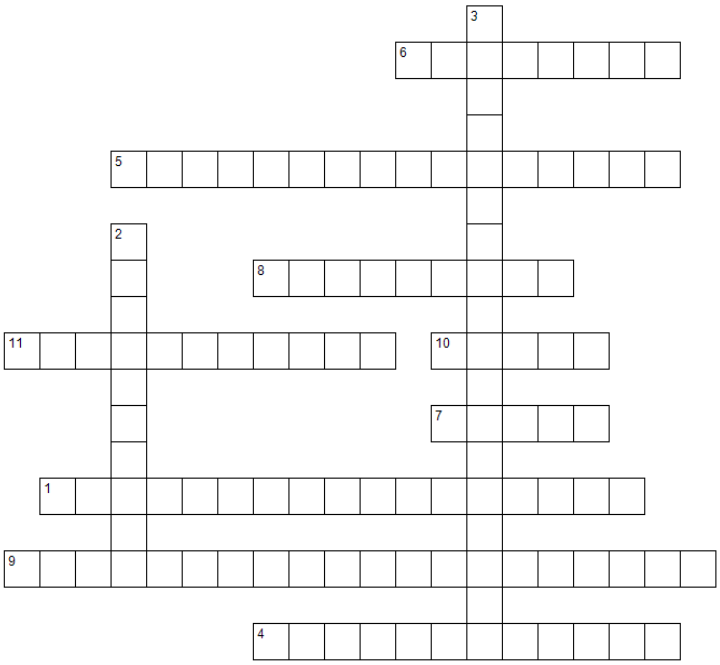
Lateral modifications cells attach tightly to one another by specialized intercellular junctions. Their functions are to establish membrane polarity, connect adjacent cells and anchor the epithelial tissue to the connective tissue underneath it. There are three types of lateral modifications:

- Tight junction (occluding junction) – act as barriers that regulate the movement of water and solutes between epithelial layers and prevent leakage of extracellular fluid
- Gap junction (communicating junction) – passageway between adjacent cells that allow the transfer of small molecules between adjacent cells
- Adhering junction – anchoring junction on the lateral surface of the cell and fasten cells to one another.

What's More

Activity 1: Uncover Me

Directions: Fill up the box with the necessary information based on the provided clues.



Across:

1. Specialization that promotes survival of organism
4. Transfer of small molecules between adjacent cells
5. Transport fluid
6. Microscopic hair-like organelle used for movement
7. Short, hair-like projections
8. Act like spot welds
9. Apical surface
10. Finger-like projections
11. Long, non-motile microvilli

Down:

2. FInger-like cytoplasmic projections
3. Basal surface

What I Have Learned

Think of this, how does cell modifications play a vital role to cells' processes and functions?

_____

_____

What I Can Do

Metastasis is the spread of cancer cells to tissues and organs beyond where the tumor originated, and the formation of new tumors (secondary and tertiary foci) is the single event that results in the death of most patients with cancer. Explain the role of tight junctions (TJ) in preventing metastasis.

Lesson 6 — Cell Cycle and Its Checkpoints

Cell cycle, a series of growth and development steps a cell undergoes between its "birth"—formation by the division of a mother cell—and reproduction—division to make two new daughter cells. It is a 4-stage process consisting of Gap 1 (G1), Synthesis, Gap 2 (G2) and Mitosis. After completing the cycle, the cell either starts the process again from G1 or exits the cycle through G0. From G0, the cell can undergo terminal differentiation. *In eukaryotic cell, or cell with nucleus, cell cycle has two stages: interphase and mitotic phase.*

What's In

Actively dividing eukaryote cells pass through a series of stages known collectively as the cell cycle: two gap phases (G1 and G2); an S (for synthesis) phase, in which the genetic material is duplicated; and an M phase, in which mitosis partitions the genetic material and the cell divides. The period between mitotic divisions - that is, G1, S and G2 - is known as interphase.

G1 phase. Metabolic changes prepare the cell for division. At a certain point - the restriction point - the cell is committed to division and moves into the S phase.

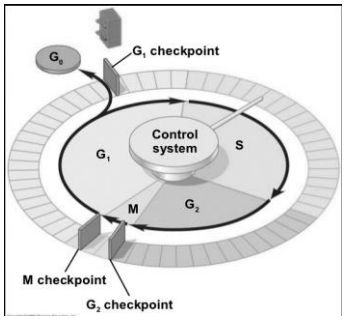
S phase. DNA synthesis replicates the genetic material. Each chromosome now consists of two sister chromatids.

G2 phase. Metabolic changes assemble the cytoplasmic materials necessary for mitosis and cytokinesis.

M phase. A nuclear division (mitosis) followed by a cell division (cytokinesis).

What's New

A **checkpoin**t is one of several points in the eukaryotic cell cycle at which the progression of a cell to the next stage in the cycle can be halted until conditions are favorable (e.g. the DNA is repaired). These checkpoints occur near the end of G_1, at the G_2/M transition, and during metaphase.



What is It

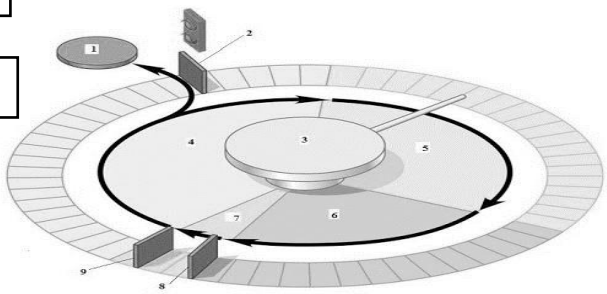
Internal Checkpoints

- Damage to DNA and other external factors are evaluated at the **G1** checkpoint (restriction point); if conditions are inadequate, the cell will not be allowed to continue to the **S** phase of interphase.
- The **G2** checkpoint ensures all of the chromosomes have been replicated and that the replicated DNA is not damaged before cell enters mitosis.
- The **M** checkpoint (spindle point) determines whether all the sister chromatids are correctly attached to the spindle microtubules before the cell enters the irreversible anaphase stage.

What's More

Directions: Label the figure on the side to test your knowledge about stages of cell cycle. Describe the events that took place in each stage.

Control System | G_2 Checkpoint | G_1 Checkpoint | M Checkpoint | G_2 | G_1 | M | G_0 | S



What I Have Learned

Directions: Based on what you have learned from this lesson, write a concise explanation of the following questions.

1. What is cell cycle checkpoints?

_____

2. How are G1, M, and G2 Checkpoints differ from each other?

_____

What I Can Do

Why do you think it is important for a cell to grow before it replicates its DNA? How important is check points in the cell cycle to organisms?

_____

_____

Assessment

Directions: Read each situation carefully. Write the letter of the correct answer on the space provided.

____ 1. When during the cell cycle are chromosomes visible?
A. only during interphase C. only during cell division
B. only when they are being replicated D. only during the G1 phase

____ 2. During which phase in the cell cycle does mitosis happen?
A. G1 phase B. G2 phase C. M phase D. S phase

____ 3. Which pair includes a phase of the cell cycle and a cellular process that occurs during that phase?
A. G1 phase, DNA replication C. S phase, cell division
B. G2 phase, preparation for mitosis D. M phase, cell growth

____ 4. At which phase during the cell cycle is cell's DNA replicated?
A. G1 phase B. G2 phase C. M phase D. S phase

____ 5. Which of the following is a correct statement about the events of the cell cycle?
A. Little happens during the G1 and G2 phases.
B. DNA replicates during cytokinesis. .
C. The M phase is usually the longest phase
D. Interphase consists of the G1, S, and G2 phases

Additional Activities

Using the recyclable materials found in your house; create a representation of cell cycle with its checkpoints guided by a rubric provided below.

	Excellent (10)	**Satisfactory (9-8)**	**Needs Improvement (6)**	**Unacceptable (3)**
Appearance	Project stands out from the rest, shows evidence of **considerable** effort.	Appearance is neat, **labels are typed or neatly written**, project is organized, and shows evidence of good effort.	Labels are hard to read, project is not neatly done, more effort needed	Appears hastily built, lack of effort is evident.

Creativity	Cell model uses materials not often seen in most projects.	Good, creative use of materials that are often used by other students.	Minimal creativity is used; **project is a poster or other 2-D model.**	Lacks creativity, copied from diagram in book.
Model Type	Cell model **clearly** represents a cell cycle	Cell model represents somewhat cell cycle	Model is a replica of a generalized sample	
Details	All parts are accurately detailed and clearly represented. Actual numbers of phases are represented.	Parts are accurately detailed and clearly recognizable. Actual numbers of phase are represented.	More detail needed to recognize cell cycle parts. Some are not recognizable. Numbers of cell cycle parts are somewhat representative of an actual cell.	Parts of cell are generalized "blobs" of color. Numbers of phases are NOT representative of an actual cell.
Labeled Parts	11+ parts are correctly located and labeled on the model.	8-10 parts are correctly located and labeled on the model.	5-7 parts are correctly located and labeled on the model.	<5 parts are labeled OR there are **errors** with parts identified.
Events are described	Functions of all events are correctly described in **detail**.	Functions of all events are correctly summarized.	Functions of some events are summarized with minor errors.	Functions are not clearly explained or **contain errors**.

Lesson 7 — Cell Division

Cell division is the process cells go through to divide. There are several types of cell division, depending upon what type of organism is dividing. Organisms have evolved over time to have different and more complex forms of cell division. Most prokaryotes, or bacteria, use binary fission to divide the cell. Eukaryotes of all sizes use *mitosis* to divide. Sexually reproducing eukaryotes use a special form of cell division called *meiosis* to reduce the genetic content in the cell. This is necessary in sexual reproduction because each parent must give only half of the required genetic material, otherwise the offspring would have too much DNA, which can be a problem. These different types of cell division are discussed below.

Learning Competencies:

- Describe the stages of mitosis/meiosis given 2n=6
- Explain the significance or applications of mitosis/meiosis

What I Know

Directions: Fill in the K-W-H-L Chart below to assess your prior knowledge and understanding of the topic.

K	W	H	L
What do I know?	**What do I want to find out?**	**How can I found out what I learn?**	**What did I learn?**

What's In

Directions: Unscramble the letters to come up with the correct words.

1. SISMITO _____
2. EIOSIMS _____
3. EMATASEHP _____
4. HASEPPRO_____
5. NAAPHSEA _____
6. TLOEPASEH_____
7. KINESISCYTO_____
8. EGNEISSOO_____
9. PERMSATOEESISG_____
10. ARKYOKINESIS _____

What's New

There are two types of cell division that cells may undergo- mitosis and meiosis. Mitosis is a type of cell division that happens in your somatic cells or body cells, in which each daughter cell receives the same number of chromosomes as the parent cell. Each daughter cell contains 46 (2n) chromosomes as the parent cell. It involves in growth, repair, and asexual reproduction. It involves four stages: prophase, metaphase, anaphase, and telophase (PMAT). The process is summarized below:

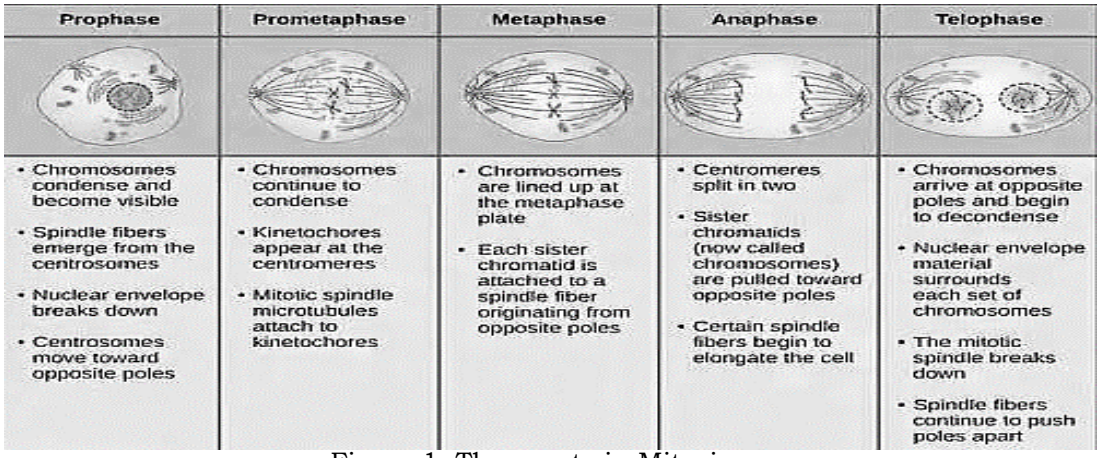


Figure 1. The events in Mitosis

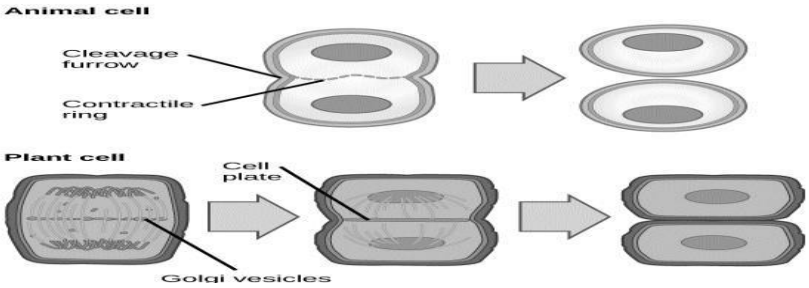


Figure 2. Comparison of mitosis in plant cell and animal cell

What is It

As you reached puberty, some somatic cells start to mature and develop into sex cells because of the hormonal changes of the sex hormones produced by your reproductive system: estrogen for females and testosterone for males. This process of producing sex cells is called gametogenesis. Your reproductive cells- the sperm and egg cells – are form **i**n sexually reproducing animals, it is usually necessary to reduce the genetic information before fertilization. Some plants can exist with too many copies of the genetic code, but in most organisms it is highly detrimental to have too many copies. Humans with even one extra copy of one chromosome can experience detrimental changes to their body. To counteract this, sexually reproducing organisms undergo a type of cell division known as *meiosis*. As before mitosis, the DNA and organelles are replicated. The process of meiosis contains two different cell divisions, which happen back-to-back. The first meiosis, *meiosis I*, separates homologous chromosomes. The homologous chromosomes present in a cells represent the two alleles of each gene an organism has. These alleles are recombined and separated, so the resulting daughter cells have only one allele for each gene, and no homologous pairs of chromosomes. The second division, *meiosis II*, separated the two copies of DNA, much like in mitosis. The result of meiosis in one cell is 4 cells, each with only one copy of the genome, which is half the normal number. ed through another kind of cell division called meiosis.

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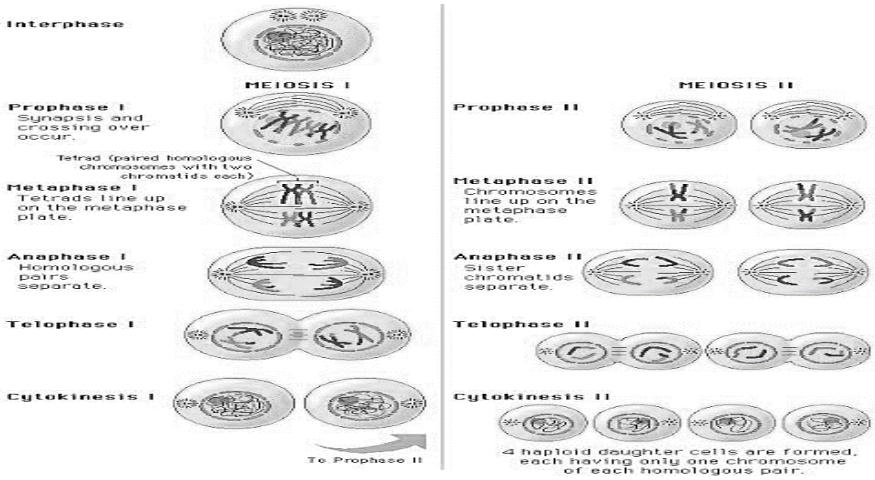
What's More

Activity 1: Divide-Reduce

Directions: Analyze the diagram and answer the guide questions below.



https://www.google.com/search?q=phschool.com-+mitosis+%26+meiosis&tbm

Figure 3 The comparison of Meiosis I and Meiosis II

Guided Questions:

1. What structure in a cell's nucleus holds the hereditary information? _____
2. How many daughter cells are produced from each mitotic cell division? _____
3. How many chromosomes does each daughter cell have after the Meiosis I? Meiosis II? _____

What I Have Learned

Directions. Complete the table.

Basic Comparison	**MITOSIS**	**MEIOSIS**
Kind of cell where the process happens		
Number of cell divisions		
Number of chromosomes in parent cell		
Number of chromosomes in daughter cell		

What I Can Do

Why is cell division important for the growth of organism? _____
_____.

Assessment

Multiple Choice: Choose the letter of the best answer. Write your answer on the space provided before the number.

_____1. In which stage of meiosis do synapsis and crossing-over occur?
A. Anaphase I B. Interphase I C. Metaphase II. D. Prophase II

_____2. Which of the following cells undergo meiosis?
A. cardiac muscle cells C. skin cells
B. red blood cells D. sperm cell

_____3. Meiosis is a type of cell division that produces_____
A. chromosome B. DNA. C. sex cells D. zygote

_____4. What happens to daughter cells after meiosis?
A. They are duplicated. C. They are diploid (2n)
B. They are haploid (n) D. They become larger in size

_____5. Crossing over means_____
A. complete meiosis.
B. produces haploid cells.
C. continues with meiosis II
D. makes genetic recombination possible

_____6. If a cell has 24 at the beginning of meiosis, how many chromosomes will it have at the end of meiosis?
A.6 B.12 C.24 D. 48

_____7. All of the following are true of meiosis EXCEPT
A. synapsis occurs during prophase I
B. cross-over occurs during prophase I
C. in plants, spindle fibers are attached to the centriole
D. there is no replication of chromosomes between meiosis I and II

_____8. Homologous chromosomes separate during _____
A. anaphase I B. anaphase II C. prophase I D. prophase II

_____9. Human sex cells normally contain _____chromosomes.
A. 23 B.24 C. 25. D.26

_____10. The cells that would likely undergo meiosis are
A. chondrocytes B. neurons C. oogonia D. osteocytes

Additional Activities

Directions: Trace your right hand and write your pledge to your cells in keeping them healthy to avoid diseases.

Lesson 8 — Diseases and Disorders of Errors in Cell Division

Learning Competencies:

1. Identify disorders and diseases that result from the malfunction of the cell during the cell cycle. **(STEM_BIO11/12-Id-f-10)**
2. Describe the structural components of the cell membrane. **(STEM_BIO11/12-Ig-h-11)**
3. Relate the structure and composition of the cell membrane to its function. **(STEM_BIO11/12-Ig-h-12**

What I Know

Directions: Write TRUE if the statement is correct. FALSE if the statement is incorrect**.**

_____1. Trisomy is the most common genetic cause of mental retardation.
_____2. Nondisjunction is the failure of chromosome pairs to separate during cell division.
_____3. Patau Syndrome is a trisomy 21.
_____4. Edward syndrome is a trisomy 18.
_____5. Down Syndrome is a trisomy 21.

What's New

Error in cell division is generally due to failure to separate chromosomes and may result in chromosomal mutation. In some cases, some cells in an individual have a mutant version of the gene while other cells in the individual have a normal version of the same gene called Mosaicism. If the mutant genotype is widespread and harmful enough, the mutation can have a major impact on the organism. Two

examples related to mosaicism are hemophilia, a blood –clothing disorder and Marfan syndrome affects the body's connective tissues.

In meiosis, there are instances that an egg and sperm cell produce 1 copy (monosomy) or 3 copies (trisomy) instead of 2 copies of each homologous chromosome. In humans, this means either having 45 or 47 chromosomes, instead of normal 46 chromosomes.

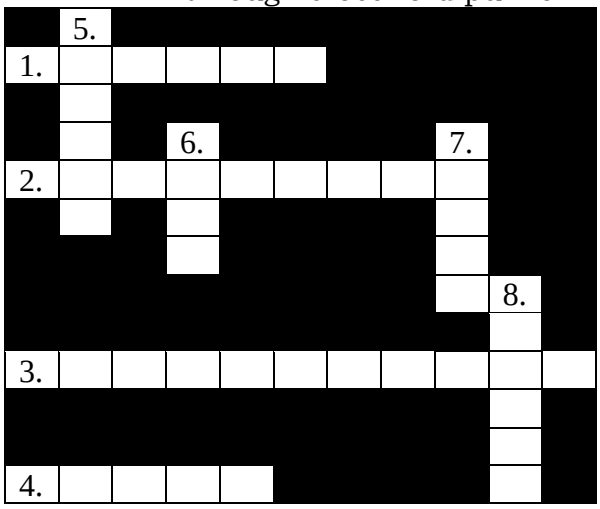
The most common disorder during meiosis is Down Syndrome, where an individual has 3 copies of chromosome 21, instead of 2. Individuals with Down Syndrome have common physical appearance such as: reduced muscle tone, eyes that slant upwards and outwards, small nose, and flat nasal bridge, single transverse palmar crease; they also have some degree of learning disability and delayed development.

Some of the trisomy disorders are: Patau Syndrome (trisomy 13) and Klinefelter's Syndrome (XXY). An example of monosomy disorders is Turner's Syndrome (XO), where an individual has only one sex chromosome X instead of XY (female) or XY (male) and Jacob's syndrome, has genetic makeup of XYY**.**

What's More

Activity 1: Hunt Time

Directions: Identify the following disorders or diseases describe in each number through crossword puzzle.



ACROSS	DOWN
1. unusual long limb	5. abnormal type of cell
2. bleeder's disease	6. have 3 copies of chromosome 21
3. have extra X (XXY)	trisomy 13
4. has genetic makeup of XYY	where an individual has only the single sex chromosome X instead of XY (female) or XY (male).

What I Have Learned

Directions. Complete the table by supplying the characteristics of the following trisomy disorders.

Trisomy Disorders	Characteristics
Down Syndrome	
Klinefelter Syndrome	

What I Can Do

What are the diseases associated with errors in mitosis?

_____

Lesson 9	Components and Functions of Plasma Membrane

What I Know

Directions: Write FACT if the statement is correct and BLUFF if the statement is incorrect.

_____1. Cell Membrane is fully permeable.
_____2. Steroid is a major component of the cell membrane.
_____3. Phospholipids is the primary building block of cell membrane.
_____4. Phospholipids composed of hydrophilic and hydrophobic head.
_____5. Hydrophobic means water loving.

What's In

The cell membrane, also called the plasma membrane provides a cellular barrier that separates cell from external environment. It protects the organelles inside the cell. It has a semi-permeable membrane that selectively controls and regulates the entry and exit of the substance. What do you think will happen if it has no cell membrane?

Even though the molecules are in constant random in motion, the cell does not permit these molecules to move in and out freely. Likewise, the cell cannot stop substances from the external environment to diffuse into the cell.

What's New

Composed of a phospholipid bilayer with embedded proteins, the plasma membrane is selectively permeable to ions and organic molecules and regulates the movement of substances in and out of cells. Plasma membranes must be very flexible in order to allow certain cells, such as red blood cells and white blood cells, to change shape as they pass through narrow capillaries.

The plasma membrane also plays a role in anchoring the cytoskeleton to provide shape to the cell, and in attaching to the extracellular matrix and other cells to help group cells together to form tissues. The membrane also maintains the cell potential.

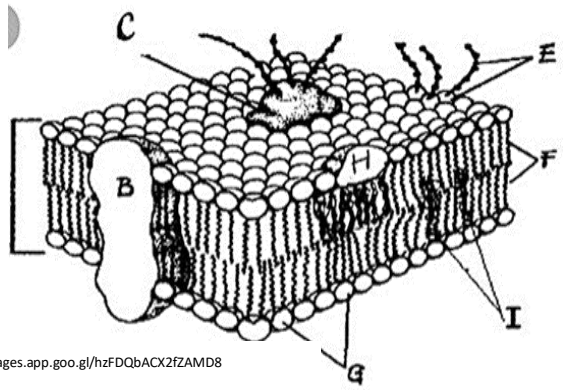
In short, if the cell is represented by a castle, the plasma membrane is the wall that provides structure for the buildings inside the wall, regulates which people leave and enter the castle, and conveys messages to and from neighboring castles. Just as a hole in the wall can be a disaster for the castle, a rupture in the plasma membrane causes the cell to lyse and die.

What's More

Activity: Color Me

Directions: Correctly color code and identify the name for each part of the cell membrane.

Phospholipid bilayer (no color)
Integral protein (orange)
Fatty acid tails (black)
Phosphate heads (yellow)
Peripheral protein (red)
Cholesterol (blue)
Glycoprotein (green)
Glycolipids (purple)



Source: http://images.app.goo.gl/hzFDQbACX2fZAMD8

What I Have Learned

Directions. Match the cell membrane structure or its function with the correct letter from the cell membrane diagram above.

_____17. Attracts water	_____21. Repels water
_____18. holds phospholipids; helps the fluidity and flexibility of the cell membrane	_____22-23) Make up the bilayer
_____19-20)Involve in cell-to-cell recognition	_____24.)Help transport certain materials across the cell membrane

What I Can Do

Make a fluid mosaic model of plasma membrane out of indigenous materials or recycle materials.

Assessment

Directions: Encircle the letter of the best answer.

1. The genetic makeup XXY corresponds to:
   A. Klinefelter's syndrome C. Tripo-X
   B. Turner Syndrome D. Jacob Syndrome
2. Which of the following terms related to Down Syndrome?
   A. Trisomy 12 B. Trisomy 13 C. Trisomy 21 D. Trisomy 31
3. Which plasma membrane component can be either found on its surface or embedded in the membrane structure?
   A. Protein B. Cholesterol C. Carbohydrates D. Phospholipids

4. What is the primarily function of the carbohydrates attached to the exterior of cell membranes?
   A. Identification of the cell
   B. Flexibility of the membrane
   C. Strengthening the membrane
   D. Channels through membrane
5. What of the following component of the cell membrane that are both hydrophilic and hydrophobic?
   A. protein B. cholesterol C. carbohydrates D. phospholipids

Additional Activities

Make a research about other diseases and disorders of errors in cell division.

Lesson 10 Transport Mechanisms in Cell

It is important for the cell to be able to transport molecules in and out of itself. Cell transport is the movement of a substance across the semipermeable, phospholipid bilayer cell membrane. A substance sometime needs the assistance of a protein, like channel protein or a transmembrane protein to cross the cell membrane. In this way, homeostasis state within cells, tissues, organs, and organ systems of a body stays alive and healthy.

Learning Competency:

- Explain transport mechanisms in cells. **(STEM_BIO11/12-Ig-h-13)**

What I Know

Directions: Read and analyze the following statements. Write the letter of the correct answer on the space provided.

_____1. What are the similarities and differences between channel proteins and carrier proteins?
   A. Both form continuous protein pathways across the phospholipid bilayer.
   B. Both have receptors that bind to a specific molecule (substrate) needing transport.
   C. Both can transport ions and molecules passively through facilitated diffusion.
   D. Both can perform in an active transport only.

_____2. What are the differences between channel proteins and carrier proteins?
   A. Channel proteins facilitate diffusion of molecules across the cell membrane while carrier proteins operate without energy and move molecules down their concentration gradient.
   B. Both can perform in a passive transport only.
   C. Both can perform in an active transport only.
   D. Carrier proteins can be either active or passive, solute flow through channel proteins is always passive.

_____3. This type of solution has the highest concentration of particles.
A. hypotonic B. isotonic C. hypertonic D. hyperactive
_____4. In facilitated diffusion, molecules move or diffuse _____.
A. against a concentration gradient
B. down a concentration gradient
C. opposed to a concentration gradient
D. reverse to a concentration gradient
_____5. It is the simplest form of passive transport.
A. diffusion B. osmosis C. facilitated diffusion D. exocytosis

What's In

Transport across cell membrane

All cells are generally separated from their surrounding environment by plasma membrane. These are semi-permeable in nature. The important role is to controls everything that enters and leaves the cell.

Substances can cross the plasma membrane in two basic ways: these are the passive transport which requires no energy and active transport which requires an energy.

Movement across a membrane and energy

There are two major ways that molecules can be moved across a membrane. The distinction has to do with whether cell energy is used. Passive mechanisms like diffusion use no energy, while active transport requires energy to get done.

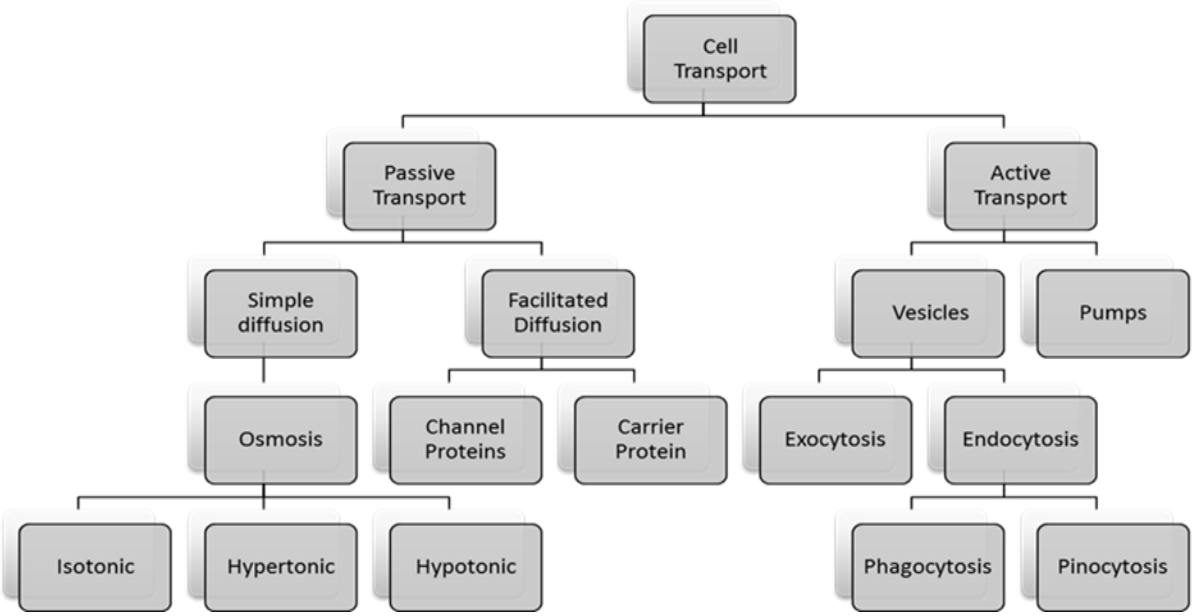


Fig. 1 The Cell Transport Concept Map illustrates various types of cell transports that happen at the plasma membrane
https://bio.libretexts.org/Bookshelves/Human_Biology/Book%3A_Human_Biology_(Wakim_and_Grewal)/05%3A_Cells/5.07%3A_Cell_Transport

What's New

Cell Transport Analogy

Passive transport is like swimming in the direction of water flow, while active transport is swimming against the water. Facilitated diffusion is like entering a subway gate using a ticket. Diffusion like the doors opening on Black Friday and the crowd rushes in the store.

What is It

A. Transport Without Energy

Passive transport occurs when substances cross the plasma membrane without any input of energy from the cell. No energy is needed because the substances are moving from an area where they have a higher concentration to an area where they have a lower concentration.

Diffusion - is defined as the net movement of molecules from an area of greater concentration to an area of lesser concentration.

1.1 Simple diffusion - is the process by which solutes move along a concentration gradient **a** solution or across a semi-permeable membrane.

1.2 Osmosis - the movement of water through a semi-permeable membrane from a region of high concentration to a region of low concentration, tending to equalize the concentrations of the water.

These are three possible relationships that cells can encounter when placed into a sugar solution.

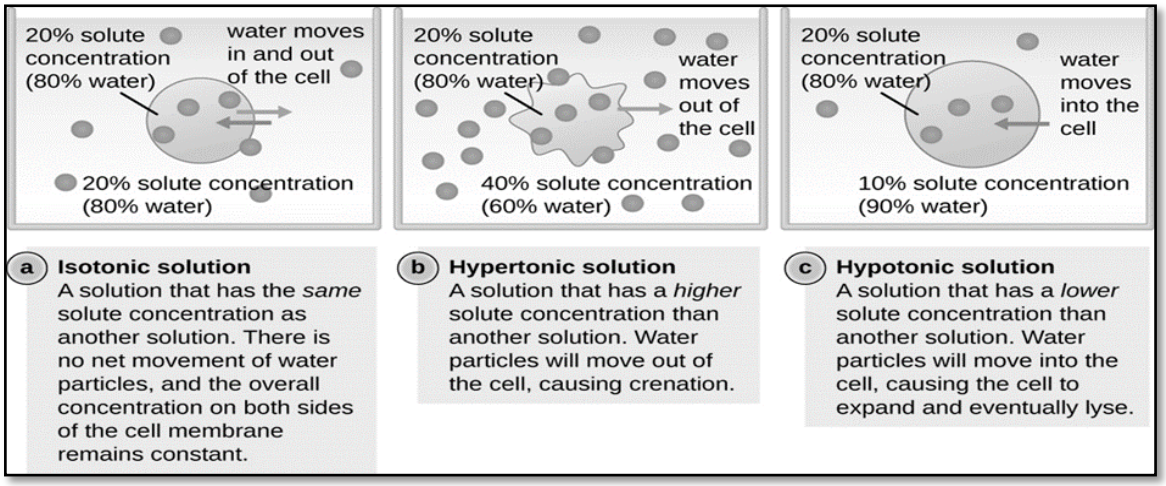


Fig. 2: Osmosis through the semi-permeable membrane of the cells. (CC BY 4.0; OpenStax via Wikimedia.org

1.2 Facilitated Diffusion - is a protein-facilitated movement of solutes across a membrane from an area of higher concentration to an area of lower concentration.

1.2.a. Channel proteins - form pores, or tiny holes, in the membrane. This allows water molecules and small ions to pass through the membrane without encountering the hydrophobic tails of the lipid molecules in the interior of the membrane.

1.2.b. Carrier proteins - bind with specific ions or molecules, and in doing so, they change shape. As carrier proteins change shape, they carry the ions or molecules across the membrane. Both protein molecules help substances diffuse across a cell membrane.

B. Transport with the Use of Energy

2. Active transport - the processes of moving materials through the cell membrane that requires the use of energy.

2.1. Pumps - usually happens across the cell membrane, proteins must work against a concentration gradient. That term means they are pumping something (usually ions) from areas of lower to higher concentration. The sodium- potassium pump moves sodium ions out of and potassium ions into the cell.

2.2. Vesicles -Membrane-bound and secreted proteins are made on ribosomes found in the rough endoplasmic reticulum. Can move molecules between locations inside the cell, e.g., proteins from the rough endoplasmic reticulum to the Golgi apparatus.

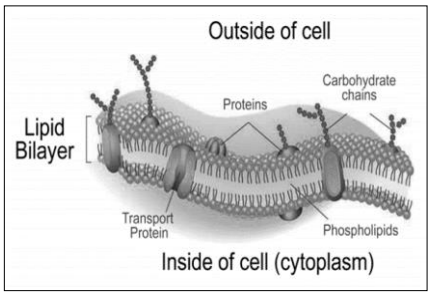


Fig. 5 Structure of the cell membrane
https://medium.com/@biologynotes/cell-membrane-dc52e97b808b

2.2.a. Exocytosis - a process by which the contents of a cell vacuole are released to the exterior through fusion of the vacuole membrane with the cell membrane.

2.2.b. Endocytosis - is the process of actively transporting molecules into the cell by engulfing it with its membrane.

2.2. b.1. Phagocytosis – is a process wherein a cell binds to the item it wants to engulf on the cell surface and draws the item inward while engulfing around it. It happens when the immune system cell is trying to destroy something, like a virus or an infected cell.

2.2. b.2. Pinocytosis - is a cellular process by which fluids and nutrients are ingested by cells. Cell drinking, pinocytosis is a type of endocytosis that involves the inward folding of the cell membrane (plasma membrane) and the formation of membrane-bound, fluid-filled vesicles.

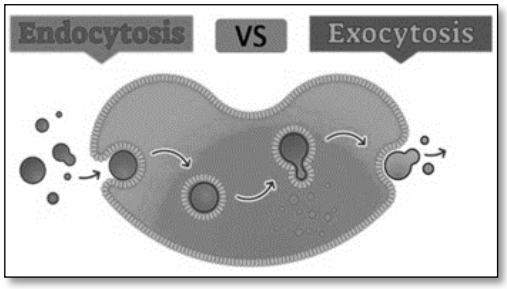


Fig. 3 Endocytosis and exocytosis illustration
https://microbenotes.com/endocytosis-vs-exocytosis/

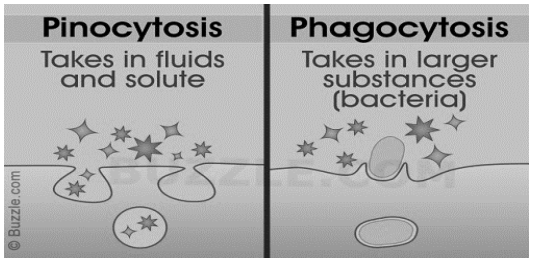


Fig. 4 Pinocytosis and phagoocytosis illustration
https://biologywise.com/difference-between-phagocytosis-pinocytosis

Watch: https://www.ck12.org/biology/passive-transport/
https://www.youtube.com/watch?v=QspmZf_yWyU
(You may use this link if you have an available internet connection, just leave it if you do not have any. This is for enrichment only)

Cell membrane architecture in transport across cell membrane

The plasma membrane plays an important role in transport of molecules. Because it acts as a semi-permeable barrier, allowing specific molecules to cross while fencing the majority of organically produced chemicals inside the cell. It consists of phospholipid bilayer (two layers of lipids) which has a polar (hydrophilic) head and two nonpolar (hydrophobic) tails. These phospholipids are aligned tail to tail so the non-polar areas form a hydrophobic region between the hydrophilic heads on the inner and outer surfaces of the membrane.

What's More

Directions: Fill in the blank of the words from the box to complete the paragraph.

simple diffusion	osmosis	passive
oxygen	facilitated diffusion	water
carbon dioxide	diffusion	blood
Lungs	heart	body
	energy	

Three common types of passive transport include_____, _____and _____.

_____ transport, in order to understand how substances, move passively across a cell membrane, it is necessary to understand concentration gradients and diffusion.

_____, _____, and _____ are among the few simple molecules that can cross the cell membrane by _____. Diffusion is one principle method of movement of substances within cells, as well as the method for essential small molecules to cross the _____.

What I Have Learned

Write your idea to complete the given key concepts.

Simple diffusion is _____ and facilitated diffusion uses _____while osmosis is_____.

What I Can Do

Vegetable vendors in market often spray water or soak their vegetables in a pale of water. Why do you think they are doing it? Which among the transport processes best describe the spraying or soaking the vegetables in water?

_____

_____.

Assessment

Directions: Read the statement carefully. Write the letter of the correct answer on the blank provided. Use capital letters.

_____1. A solution with a higher level of solute than solution is called _____.

A. hypotonic B. isotonic C. hypertonic D. hyperactive

_____2. Which process needs carrier proteins in the cell membrane?

A. passive transport C. active transport

B. carrier protein D. transport vesicle

_____3. What is the difference between active and passive transports?

A. active transport does not need energy and passive uses ATP (energy)

B. active transport stores transport proteins and passive transport releases

C. active transport uses energy (ATP) and passive transport does not need energy

D. active transport releases transport proteins and passive transport stores

_____4. A molecule that diffuses down a concentration gradient flows from an area of _____ concentration into an area of _____ concentration.

A. higher, lower
B. passive, active
C. lower, higher
D. active, passive

_____5. Why does active transport require energy?

A. To move solutes against a concentration gradient.
B. To move solutes with a concentration gradient.
C. To move solutes to an equal concentration inside and outside the cell
D. To transport small, hydrophobic molecules across plasma membrane

Additional Activities

Directions: Tabulate the differences of passive and active transports.

Passive Transport	Active Transport
1.	1.
2.	2.
3.	3.
4.	4.

Lesson 11 Components of Enzymes

Enzymes are proteins which is responsible for different metabolic processes. These proteins are produced by living cells that helps in the breakdown of food, rearrangement of certain macromolecules inside our body and/or creation of substances needed by the body.

Learning Competency

- Describe the components of an enzyme. **(STEM_BIO11/12-Ii-j-17)**

What I Know

Directions: Fill in the K-W-H-L Chart below to assess your prior knowledge and understanding of the topic, Structure and Functions of Enzymes.

K	W	H	L
What do I know?	**What do I want to find out?**	**How can I found out what I learn?**	**What did I learn?**

What's In

Each enzyme is specific for the **substrate** (a reactant that binds to an enzyme) upon which it acts. Therefore, majority of enzyme reacts on one type of substrate only. This substrate will bind to the active site of the enzyme to begin the reaction. We call enzymes that break down their substrate's **catabolic enzymes**. Those that build more complex molecules from their substrates are **anabolic enzymes**. Observe the figures below on how enzymatic reaction is taken place.

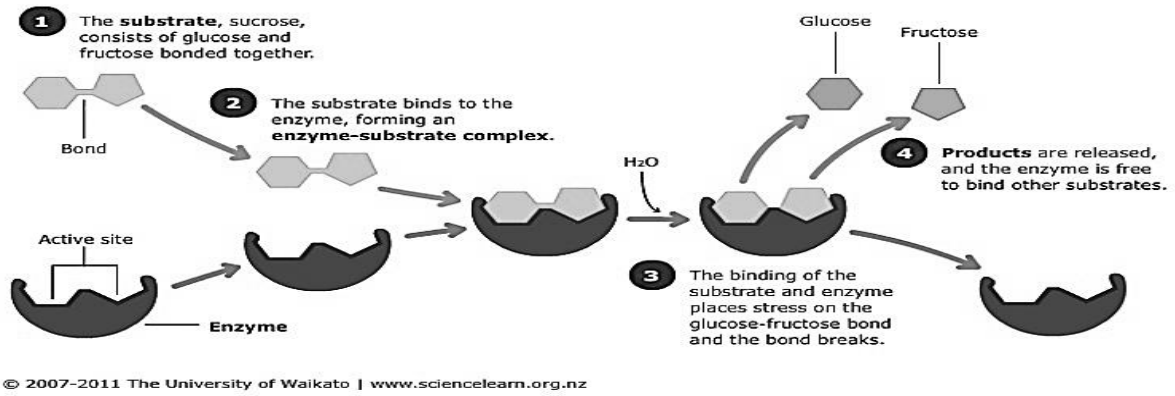


Figure 1: Catabolic reaction (breakdown molecules) via catabolic enzymes.

SOURCE:

https://www.google.com/search?q=enzyme+substrate+complex+peptide+bond&tbm=isch&ved=2ahUKEwj337vEjvbpAhWYxeAKHYpuCTAQ2cCegQIABAA&oq=enzyme+substrate+complex+peptide+bond&gs_lcp=CgNpbWcQAzoECCMQJzoCCAA6BggAEAgQHjoECAAQGFCd4gRYhYEFYOGDBWgAcAB4AIAB0gKIAfIjkgEEMy0xNJgBAKABAaoBC2d3cy13aXotaW1n&sclient=img&ei=tTngXrexE5iLgweK3aWAAw&bih=741&biw=1517&rlz=1C1CHZL_enPH724PH724#imgrc=I-4j4UCHAphDEM

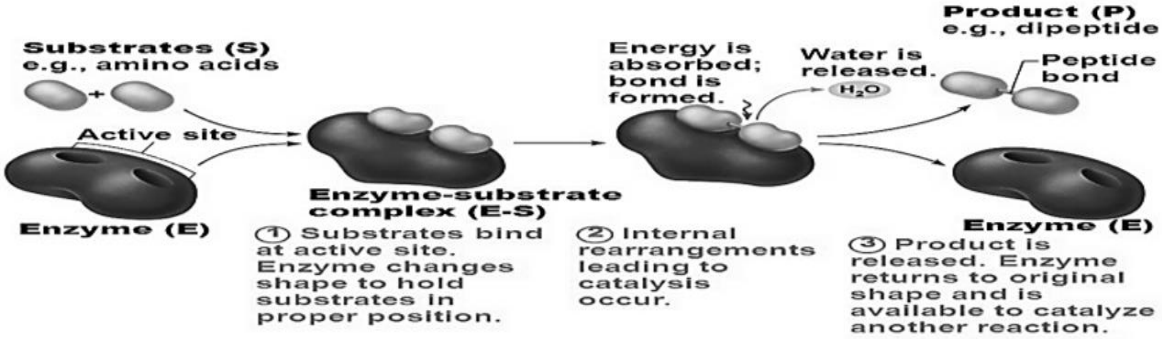


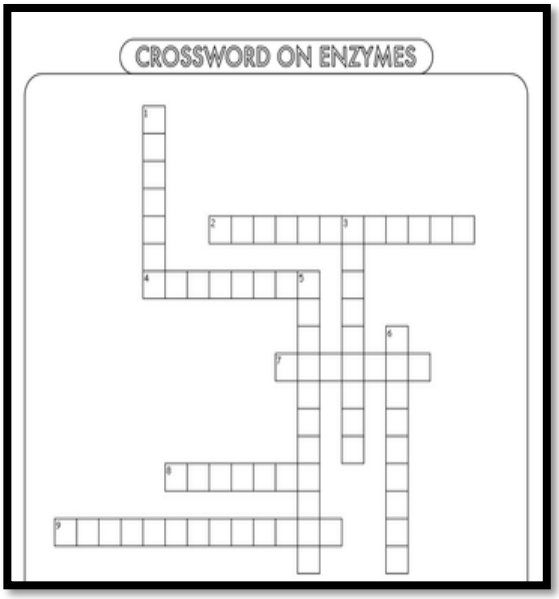
Figure 2: Anabolic reaction (builds up molecules) via anabolic enzymes.

SOURCE:

https://www.google.com/search?q=enzyme+substrate+complex+peptide+bond&tbm=isch&ved=2ahUKEwj337vEjvbpAhWYxeAKHYpuCTAQ2cCegQIABAA&oq=enzyme+substrate+complex+peptide+bond&gs_lcp=CgNpbWcQAzoECCMQJzoCCAA6BggAEAgQHjoECAAQGFCd4gRYhYEFYOGDBWgAcAB4AIAB0gKIAfIjkgEEMy0xNJgBAKABAaoBC2d3cy13aXotaW1n&sclient=img&ei=tTngXrexE5iLgweK3aWAAw&bih=741&biw=1517&rlz=1C1CHZL_enPH724PH724#imgrc=wmGKXtVA9J1o4

What's New

Directions: Read the clues and fill in the correct answer.



Across

2. The term given to when the enzyme loses its shape (12) DENATURATION
4. An enzyme is a biological one of these (8) CATALYST
7. The term used to describe the condition when the enzyme's activity is at its highest (7) OPTIMUM
8. What is released by the enzyme once the reaction has taken place (7) PRODUCT
9. The description given to how the shape of the substrate fits into the shape of the active site (13) COMPLEMENTARY

Down

1. The type of energy molecules gain as temperature increases (7) KINETIC
3. Enzymes speed these up (9) REACTIONS
5. One of the main factors which can effect enzyme activity (11) TEMPERATURE
6. The name given to the molecule that fits into the active site (9) SUBSTRATE

What is It

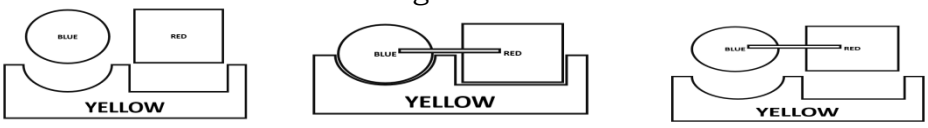
ACTIVITY 1: Team Up!

Materials:

Colored papers (red, blue, yellow)
Toothpick
short bond paper
scissors
glue

Procedure:

1. **Note**: Red and blue colored paper will represent the substrates. While yellow paper will be the enzyme.
2. Cut three (3) sets of red squared colored paper and a blue round shape.
3. Cut two (2) sets of yellow paper to create a crevice in which half of the square shaped substrate (red) and circle shaped substrate (blue) fits.
4. (Refer to figure 1 above) Using the glue attached a toothpick to the second and third sets of the enzyme substrate complex.
5. Cut out must be like the figure below.



6. Connect each shape to a short bond paper.

Guided Questions:

1. What is a substrate? _____
2. What does the crevice represent? _____
3. What do the toothpick represents in this model? _____
4. What will happened if a substrate with different shape or orientation tries to fit in to the enzyme? Briefly explain.

What's More

Directions: Explain your answer.
Why are enzymes significant in every cell and to the entire organism?

_____
_____.

What I Have Learned

Directions: Briefly explain how enzymatic reaction occurs on each situation.

1. Bananas turn brown after cutting.

   _____

2. Avocado becomes soft if it is being placed inside the rice dispenser.

   _____

What I Can Do

Directions: A. Identify if the following statements below shows ANABOLIC reactions or CATABOLIC reactions.

_____1. Combining all baking ingredients to make bread.
_____2. Amylase breaks starch into simple sugars.
_____3. Maltose synthesis found on beer.
_____4. Absorption of monosaccharide glucose and galactose from milk sugar called lactose.
_____5. Formation of proteins from amino acids.

Assessment

Directions: Identify if the following pair will have a reaction and/or action based from the concept of enzyme substrate complex. If there will be a positive reaction/action, write (/), if there's none, write (X) on the space provided before each number.

_____1. USB to HDMI port.
_____2. Door lock and key.
_____3. Laptop charger and cellphone.
_____4. Two prong plug to outlet.
_____5. Biometrics and finger print

Lesson 12	Oxidation and Reduction Reaction

All chemical reactions between molecules involve the breaking and forming of bonds. Converting starch into glucose involves contorting starch into a highly unstable state before the reaction can proceed. This unstable state is called the *transition state* that happens when reactants absorb energy from their surroundings and. This initial investment of energy in order to start a reaction is called the a*ctivation energy*.

What I Know

Directions: Identify the words being described and write it on the space provided before the number.

_____ 1. These are molecule that regulates enzyme processes.
_____ 2. Minerals are called _____ as they aid enzymatic reaction.
_____ 3. Examples of coenzymes are _____.
_____ 4. Compound that donates its electron(s) is called ____.
_____ 5. Energy required to begin chemical reaction

What's In

Oxidation–reduction reactions involve transfer of electrons between two chemical species. A compound which loses electron(s) is oxidized while the compound that gains electron(s) is reduced. A compound that donates its electron(s) is called a *reductant* or a *reducing agent*. Conversely the electron-accepting molecule is the *oxidizing agent* or *oxidant*. The oxidation and reduction events must occur together – that is, an oxidant has to be present to accept the electron(s) from a reductant. In other words, for a molecule to be reduced, some other molecule has to get oxidized.

What's New

Directions: Predict how the following factors will affect the action of an enzyme by completing the table below.

Factor affecting enzymatic reaction	Effect on reaction rate if the factor is reduced/decreased	Effect on reaction rate if the factor is raised/increased	Brief explanation as to why these factors affects enzyme function and reaction rate
Enzyme concentration			
Substrate concentration			

Inhibitor concentration			
Temperature			
pH level			

What is It

Redox reactions may be compared to proton transfers in acid–base chemistry. An acid can lose/donate a proton only when a base accepts it. No oxidant can gain electrons without another substance (reductant) losing electrons. A complete oxidation–reduction reaction is thus a combination of two half reactions (*redox couples*). When the two half reactions are combined, the component with greater tendency to gain electrons (the oxidant) gets reduced at the expense of the other (the reductant – which loses electrons).

It is often supplied as thermal energy or heat absorbed by reactants from their surroundings. Reactant molecules absorb heat which causes them to collide more frequently and more forcefully. This agitates the atoms within the molecules that results in the likely breaking of bonds. When the new bonds of the products form, energy is released as heat and the molecules return to stable shapes with lower energy. This results in an overall decrease of free energy.

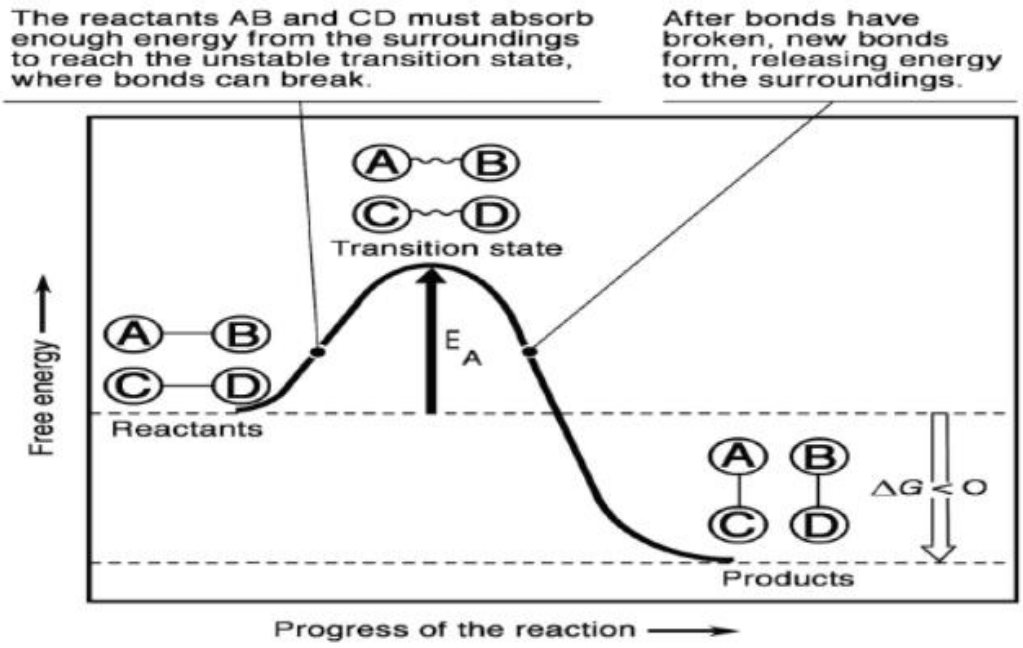


Figure 1: Enzyme change bonds of molecules

SOURCE:

https://www.google.com/search?q=Energy+profile+for+a+spontaneous+exergonic+reaction&rlz=1C1CHZL_enPH724PH724&sxsrf=ALeKk01Ab8dKUrWAOZG0hqZJNrwXxwfQww:1591754039804&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjV0vLwkfbpAhXhpHEKHR1yAwUQ_AUoAXoECA4QAw&biw=1517&bih=741#imgrc=QznvdCGXA9aPDM

Enzymatic reactions are aided by cofactors and coenzymes. Cofactors refers to inorganic molecules such as minerals. These molecules are bound within the enzyme molecule. Coenzymes are organic molecules like vitamins. These substances might bind temporarily or permanently on enzyme molecules.

What's More

ACTIVITY 2: Pair it to me gently

Materials:

Internet
Worksheet
Any available reference books

Watch a video from this web address and take down notes during the talk. **"Activation energy"**
https://www.youtube.com/watch?v=YacsIU97OFc

Guided Questions:

1. What is the activation energy of a reaction?

   _____

2. How do enzymes affect the activation energy of a reaction?

   _____

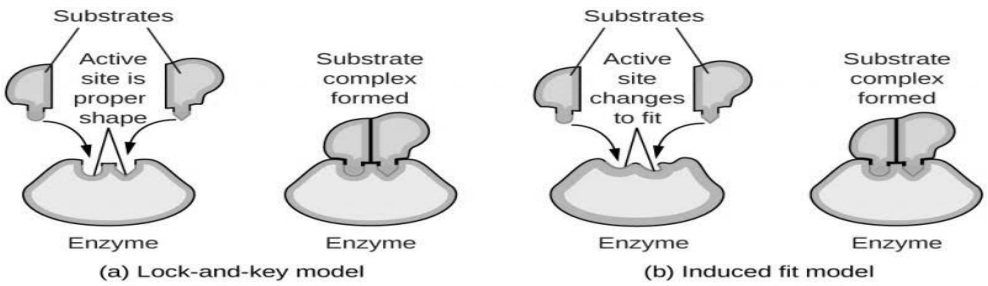
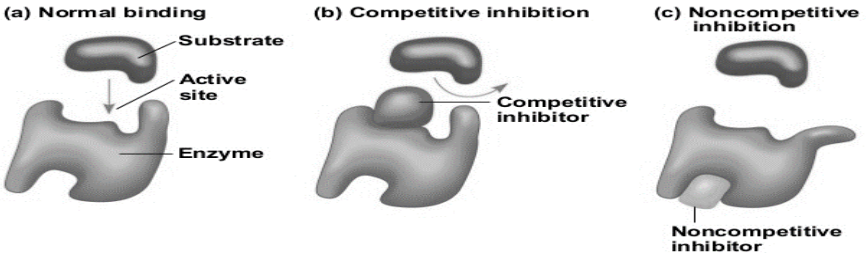


Figure 2: Enzymatic Mechanism models

ww.google.com/search?q=lock+and+key+model,+induced+fit+model&rlz=1C1CHZL_enPH724PH724&sxsrf=ALeKk03Kf-d-BrqME1huMtGaLkpUAgQ:1591752390629&source=lnms&tbm=isch&sa=X&ved=2ahUKEwi67cDei_bpAhWjonEKHUx4BNkQ_AUoAXoECAsQAw&biw=1517&bih=741&dpr=0.9#imgrc=LSFPhJnniJ1y3M



SOURCE:
https://www.google.com/search?q=types+of+enzyme+inhibitors&tbm=isch&ved=2ahUKEwju-fe1lvbpAhUI-RQKHUT5A1oQ2-cCegQIABAA&oq=types+of+enzyme+inhibitors&gs_lcp=CgNpbWcQAzICCAAyAggAMgYIABAFEB4yBggAEAUQHjIECAAQGDIECAAQGDIECAAQGDoECCMQJzoECAAQQzoGCAAQChAYUIrCAVjE1AFgnN4BaABwAHgAgAGvA4gB1BGSAQUzLTUuMZgBAKABAaoBC2d3cy13aXotaW1n&sclient=img&ei=-kHgXu6UFIjyU8Tyj9AF&bih=741&biw=1517&rlz=1C1CHZL_enPH724PH724#imgrc=snuAYUEF-saI2M&imgdii=JfunoRwLi_0UrM

Enzymatic reactions are regulated by molecules called inhibitors. Using the diagrams below, differentiate competitive and noncompetitive (allosteric) inhibition.

What I Have Learned

Enzymatic reactions are regulated by molecules called inhibitors. Using the diagrams above, differentiate competitive and noncompetitive (allosteric) inhibition. Write your answer below.

_____

_____

What I Can Do

During fever, the patient's appetite decreases so as with other metabolic processes. How does high temperature affect our body's metabolic rate, the fact humans are warm blooded species?

Assessment

Directions: Read and analyze the questions carefully. Write the letter of the best answer on the space provided

_____**1. What is the function of enzymes?**

A. to increase the ΔG of reactions
B. to increase the ΔH of reactions
C. to lower the activation energy of a reaction
D. to lower the entropy of the chemicals in the reaction

_____**2. Which of the following explains the function of allosteric inhibitor?**

A. binds to an enzyme away from the active site and changes the conformation of the active site, increasing its affinity for substrate binding
B. binds to an enzyme away from the active site and changes the conformation of the active site, decreasing its affinity for the substrate
C. binds directly to the active site and mimics the substrate
D. binds to an active site and blocks it from binding substrate

_____**3. What happens if an enzyme is not functioning in a chemical reaction in a living organism that needs it?**

A. The reaction stops.
B. There is no change in the reaction rate.
C. The reaction proceeds, but much more slowly.
D. The reaction proceeds faster without the interference.

_____**4. Which of the following is NOT true about enzymes?**

A. They increase the ΔG of reactions.
B. They are usually made of amino acids.
C. They lower the activation energy of chemical reactions.
D. Each one is specific to the particular substrate, or substrates, to which it binds.

_____**5. Which of the following analogies best describe the induced-fit model of enzyme - substrate binding?**

A. a key fitting into a lock
B. a hug between two people
C. the fitting together of two jigsaw puzzle pieces
D. a square peg fitting through the round hole of a child's toy

References

Live Science. ―Prokaryotic and Eukaryotic Cells: What ‘s the Difference?‖ Accessed on June 7, 2020. https://www.livescience.com/65922-prokaryotic-vs-eukaryotic-cells.html/graczykteach.

Prokaryotic Eukaryotic Cell Venn Diagram‖ Accessed on June 7, 2020.http://graczykteach.weebly.com/uploads/2/6/1/3/26135774/prokaryotic__eukaryotic_cells_venn_diagra m.pdf/ cobblelearning.

Prokaryote and Eukaryote Cells‖ Accessed on June 7 2020. https://cpb-se1.wpmucdn.com/cobblearning.net/dist/3/4046/files/2019/08/Prokaryote-and-Eukaryote-Cells-POGILKEY.pdf/ twinkl.

Specialized Cells‖. Accessed on June 7. 2020. https://www.twinkl.co.uk/resource/t3-sc-484-ks3-specialised-cells-activity-sheets./ stileapp.

Specialized Cells: Build your own body parts‖ Accessed on June 7, 2020.http://stileapp.com/static/CLL%20handouts/Lesson_050_handout.pdf /

Live Science. ―Prokaryotic and Eukaryotic Cells: What ‘s the Difference? ‖ Accessed on June 7, 2020. https://www.livescience.com/65922-prokaryotic-vs-eukaryotic-cells.html/ graczykteach.

Prokaryotic Eukaryotic Cell Venn Diagram‖ Accessed on June 7, 2020.http://graczykteach.weebly.com/uploads/2/6/1/3/26135774/prokaryotic__eukaryotic_cells_venn_diagra m.pdf/ cobblelearning.

Prokaryote and Eukaryote Cells‖ Accessed on June 7 2020. https://cpb-se1.wpmucdn.com/cobblearning.net/dist/3/4046/files/2019/08/Prokaryote-and-Eukaryote-Cells-POGILKEY.pdf/ twinkl.

Specialized Cells‖. Accessed on June 7. 2020. https://www.twinkl.co.uk/resource/t3-sc-484-ks3-specialised-cells-activity-sheets./ stileapp.

Specialized Cells: Build your own body parts‖ Accessed on June 7, 2020.http://stileapp.com/static/CLL%20handouts/Lesson_050_handout.pdf /

Boundless. “Boundless Biology.” Lumen. Accessed June 12, 2020. https://courses.lumenlearning.com/boundless-biology/chapter/control-of-the-cell-cycle/.

Vaskovic, Jana. “Overview and Types of Epithelial Tissue.” Kenhub. Kenhub, May 15, 2020. https://www.kenhub.com/en/library/anatomy/overview-and-types-of-epithelial-tissue.

Lecturio. “Surface Modification of Epithelial Cell Membrane and Trans-Epithelial Potential.” The Lecturio Online Medical Library. Lecturio, November 18, 2019. https://www.lecturio.com/magazine/surface-modification-epithelial-cell-membrane-and-trans-epithelial-potential/.

Google Search. Google. Accessed June 12, 2020. https://www.google.com/search?q=flagella+picture.

Google Search. Google. Accessed June 12, 2020. https://www.google.com/search?q=cilia+picture.

Google Search. Google. Accessed June 12, 2020. https://www.google.com/search?q=stereocilia+picture.

Google Search. Google. Accessed June 12, 2020. https://www.google.com/search?q=microvilli+picture.

Google Search. Google. Accessed June 12, 2020. https://www.google.com/search?q=cell+modification.

Google Search. Google. Accessed June 12, 2020. https://www.google.com/search?q=biology+3d+cell+cycle.

Google Search. Google. Accessed June 12, 2020. https://www.google.com/search?q=labeled+cell+cycle+diagram+with+checkpoints.

“Cell Cycle Stages Worksheet - 1.Cdn.edl.io.” Accessed June 12, 2020. https://1.cdn.edl.io/ZnnLOJXnciuYM5eAuaX7yKsKfmMaKKl3fijB0L1XDYOAYSAy.pdf.

“Cell Cycle Stages Worksheet.” Google Sites. Accessed June 12, 2020. https://sites.google.com/view/vqpbmvsuit/cell-cycle-stages-worksheet.

Capco, Carmelita M. *You and the Natural World: Biology*. Quezon City: Phoenix Publishing House, Inc.,2011

Campbell, Neil A.,and Jane Reece. *Biology*, 9th ed., San Francisco, California Pearson, Benjamin Cummings,2009.

Evangelista, Eden and Malonzo Edith, Science in Today ‘s World: Biology. Quezon City: Sibs Publishing House, 2008.

Youngenome Org. ―Stages of Mitosis‖. Accessed on May 30, 2020. http://www.youngenome.org/facts/what-is-mitosis

SeatlePi. Biological Significance of Mitosis & Meiosis in Sexual Reproduction. / Accessed on June 1,2020. https://education.seattlepi.com/biological-significance-mitosis-meiosis-sexual-reproduction-

Cancer Treatment Center of Asia. Accessed on June 5,2020. https://www.cancercenter.com/what-is-cancer?.

Fine, Jacob 2019. Meiosis Mistakes. http://letstalkscience.ca/educational-resources/stem-in-contetx/meiosis-mistakes

Holland, Andrew, Ph.D 2015. Tomorrows Discoveries: When Cell Division Goes Wrong. http://www.youtube.com/watch?v=6k_Y-D4Vg7A&feature=share

“Cell Transport.” Biology - Home. Accessed July 14, 2020. http://mybioscienceproject.weebly.com/cell-transport.html.

CK-12 Foundation. “12 Foundation.” CK. CK-12 Foundation. Accessed July 14, 2020. https://www.ck12.org/biology/passive-transport/

Pearson - The Biology Place. Accessed July 14, 2020. http://www.phschool.com/science/biology_place/biocoach/biomembrane1/regions.html

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